

Deterministic Optimization

Discrete Optimization:
Modeling w/ binary variables

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Nonconvex Sets

Modeling Nonconvex Sets

Learning Objectives

- Observe how to model some nonconvex sets using mixed integer **linear** constraints

Discrete Variables

- A discrete variable has a domain of the form

$$x \in \{a_1, a_2, \dots, a_K\}$$

- If $a_1, a_2, \dots, a_K$ are contiguous integers, we can model as

$$a_1 \leq x \leq a_K, \quad x \in \mathbb{Z}$$

- Otherwise we can model as

$$x = \sum_{k=1}^K a_k y_k, \quad \sum_{k=1}^K y_k = 1, \quad y_k \in \{0, 1\} \quad \forall k = 1, \dots, K$$

Semicontinuous Variables

- A semicontinuous variable has a domain of the form

$$x \in [\ell_1, u_1] \cup [\ell_2, u_2] \cup \cdots \cup [\ell_K, u_K]$$

where $\ell_1 \leq u_1 \leq \ell_2 \leq u_2 \leq \cdots \leq \ell_K \leq u_K$.

- One modeling approach for this system is

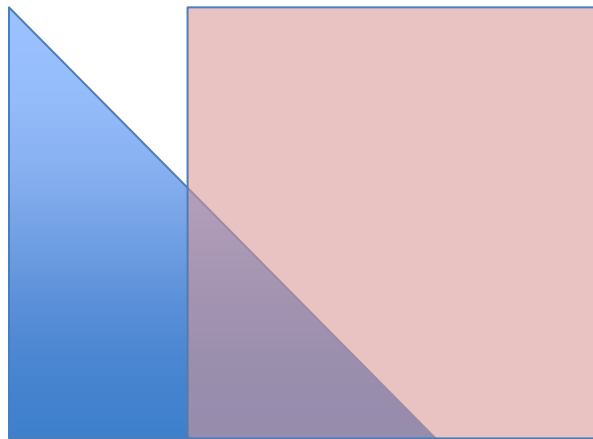
$$x = \sum_{k=1}^K z_k, \quad \ell_k y_k \leq z_k \leq u_k y_k \quad \forall k = 1, \dots, K$$
$$\sum_{k=1}^K y_k = 1, \quad y_k \in \{0, 1\} \quad \forall k = 1, \dots, K$$

Union of Polytopes

- Suppose we require that $\mathbf{x}$ satisfies **one** of the following K systems of inequalities: $\mathbf{A}_k \mathbf{x} \leq \mathbf{b}_k$ for $k = 1, \dots, K$
- Suppose that each system defines a polytope, i.e. a bounded polyhedron
- Then a formulation for this system is

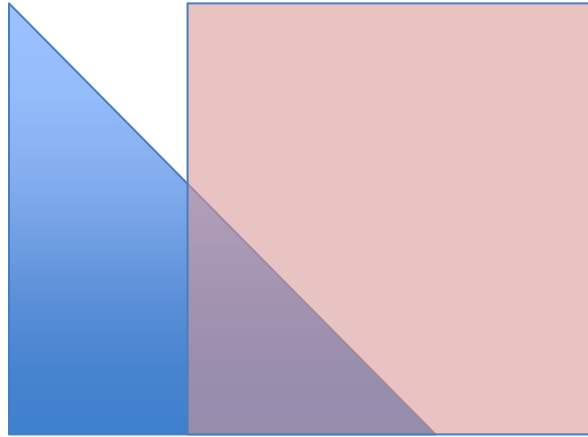
$$\mathbf{A}_k \mathbf{u}^k \leq \mathbf{b}_k y_k, \quad \mathbf{x} = \sum_{k=1}^K \mathbf{u}^k, \quad \sum_{k=1}^K y_k = 1, \quad y_k \in \{0, 1\} \quad \forall k = 1, \dots, K$$

Example 1



$$\left(\begin{array}{l} x_1 + x_2 \leq 2 \\ x_1 \geq 0, x_2 \geq 0 \end{array} \right) \cup \left(\begin{array}{l} 1 \leq x_1 \leq 3 \\ 0 \leq x_2 \leq 2 \end{array} \right)$$

Example 1

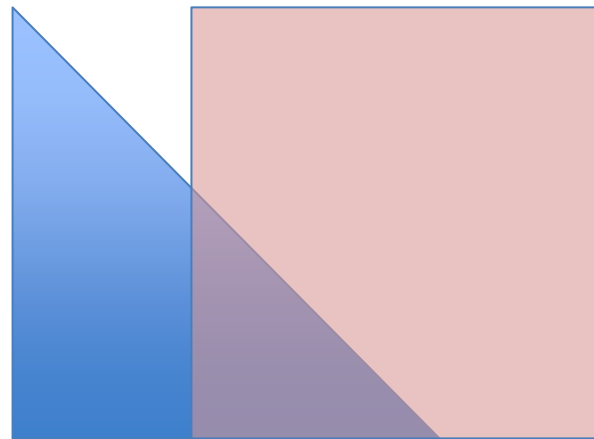


$$\left(\begin{array}{l} x_1 + x_2 \leq 2 \\ -x_1 \leq 0, -x_2 \leq 0 \end{array} \right) \cup \left(\begin{array}{l} -x_1 \leq -1, x_1 \leq 3 \\ -x_2 \leq 0, x_2 \leq 2 \end{array} \right)$$

Example 1

$$\left(\begin{array}{l} x_1 + x_2 \leq 2 \\ -x_1 \leq 0, -x_2 \leq 0 \end{array} \right) \cup \left(\begin{array}{l} -x_1 \leq -1, x_1 \leq 3 \\ -x_2 \leq 0, x_2 \leq 2 \end{array} \right)$$

$$\begin{aligned} x_1 &= u_1^1 + u_1^2, \quad x_2 = u_2^1 + u_2^2 \\ u_1^1 + u_2^1 &\leq 2y_1, \quad -u_1^1 \leq 0, \quad -u_2^1 \leq 0 \\ -u_1^2 &\leq -1y_2, \quad -u_1^2 \leq 3y_2, \quad -u_2^2 \leq 0, \quad u_2^2 \leq 2y_2 \\ y_1 + y_2 &= 1, \quad y_1 \in \{0, 1\}, \quad y_2 \in \{0, 1\} \end{aligned}$$



Bounds

- Note that we have been making use of the bounds on the variables to develop these formulations
- In general, it is impossible to develop correct mixed integer linear formulations without bounds on variables
- If there are no bounds, we will impose large artificial numbers (big-M) as variable bounds

Example 2

Model the nonconvex set $X = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\|_0 \leq k\}$ where k is an integer between 0 and n . Here $\|\mathbf{x}\|_0$ is the number of nonzero components of $\mathbf{x}$.

This set is not bounded. We will impose bounds, $-M \leq x_j \leq M$ for all $j = 1, \dots, n$ where M is a large number.

Then a mixed integer linear formulation is

$$-My_j \leq x_j \leq My_j, \quad \sum_{j=1}^n y_j \leq k, \quad y_j \in \{0, 1\} \quad \forall j = 1, \dots, n$$

Summary

- We saw some examples on how to model nonconvex sets by mixed integer linear programming
- It is important that the sets to be modeled are **bounded**
- It is very important to make sure the resulting formulations are linear